

Class 3: Boolean Algebra

Goals:

Learn terminology and form of common Boolean algebra laws.

Use Boolean Algebra laws to simplify complex statements into other statements that are logically equivalent.

Determine if a set of connectives is adequate to create every possible column in a truth table.

Definition: A Boolean algebra is

Basic Algebra Laws of Sentential Logic

Formal Name	Logical Equivalence
Double negation	$\sim (\sim p) \equiv p$
De Morgan's laws	$\sim (p \wedge q) \equiv (\sim p) \vee (\sim q)$ $\sim (p \vee q) \equiv (\sim p) \wedge (\sim q)$
Implication expansion	$p \rightarrow q \equiv (\sim p) \vee q$
Contrapositive	$p \rightarrow q \equiv (\sim q) \rightarrow (\sim p)$
Biconditional expansion	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
Commutativity	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associativity	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributivity	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Tautology	$p \vee (\sim p) \equiv T$
Contradiction	$p \wedge (\sim p) \equiv F$
Simplification	$p \wedge T \equiv p$ and $p \wedge F \equiv F$ $p \vee T \equiv T$ and $p \vee F \equiv p$

All of these basic laws are proven using truth tables by showing that the values for the two expressions are the same in all rows.

Use a truth table to prove the biconditional expansion law.

P	Q	$P \leftrightarrow Q$	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$

Use the laws in the table to simplify the expression

$$[p \wedge (q \wedge r)] \vee [(p \wedge q) \wedge (\sim r)]$$

Big Question: Is it possible to construct a logical sentence for any given truth column in a table?

How many different columns could there be?

Adequate Set of Connectives

Find a sentence involving p , q , and r that is equivalent to each target using only the connectives $\sim$, $\wedge$, $\vee$.

p	q	r	Target A	Target B	Target C	Target D	Yours
T	T	T	F	T	T	T	
T	T	F	F	T	T	F	
T	F	T	F	F	F	F	
T	F	F	T	F	T	T	
F	T	T	F	F	F	F	
F	T	F	F	F	F	F	
F	F	T	F	F	F	T	
F	F	F	F	F	F	F	

Sentence A:

Sentence B:

Sentence C:

Sentence D:

Yours:

Can you formulate a strategy to hit any target?

Disjunctive Normal Form: A disjunction of sentences where each sentence is a conjunction of sentence symbols, sentence variables, and negations.

Big Answer: It is possible to construct a logical sentence for any given truth column. Furthermore we only need the connectives $\sim$, $\wedge$, and $\vee$ to do it.

We say the set $\{\sim, \wedge, \vee\}$ is an adequate set of connectives to construct a sentence for any given truth column.

Are there other adequate sets?

Prove that $\{\sim, \wedge\}$ is an adequate set.

Known adequate set
 $\{\sim, \wedge, \vee\}$

Conjectured adequate set
 $\{\sim, \wedge\}$

Prove that $\{\sim, \rightarrow\}$ is an adequate set, given that $\{\sim, \wedge\}$ is an adequate set.